

COMSATS UNIVERSITY ISLAMABAD

Department of Computer Science

Lab Assignment 1

WEB TECHNOLOGIES

Project: COMSATS Lost & Found Portal

A Campus Lost-and-Found Recovery Web Application

Muhammad Umar Saeed (Reg. No. not yet assigned — transfer case in process)

Abdullah Shahid (FA24-BCS-139)

1. Project Overview

The COMSATS Lost & Found Portal is a campus utility web application built to help students report, browse, and recover lost or found items around the COMSATS campus. It was implemented following the Web Development lab manual's requirements for planning a website (sitemap and wireframe), building it with HTML, CSS, and JavaScript, and then testing it through a local web server before deployment.

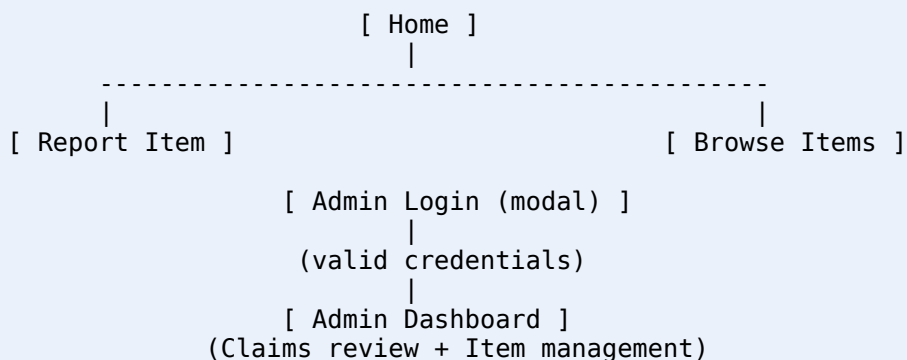
The application consists of four main views:

- Home — landing page with a project introduction, a 3-step “How It Works” explainer, and a live feed of recently reported items
- Report Item — a form for students to submit a lost or found item report (name, roll number, item type, location, title, description, contact email)
- Browse Items — a searchable, filterable listing of all reported items (by type and status)
- Admin Dashboard — a protected panel, reached via an “Admin Login” modal, where an administrator reviews statistics, claims, and manages all reported items (resolve / delete)

All pages share a single consistent dark-themed layout, a common navigation bar (Home / Report Item / Browse Items), and an Admin Login control that gates access to the dashboard.

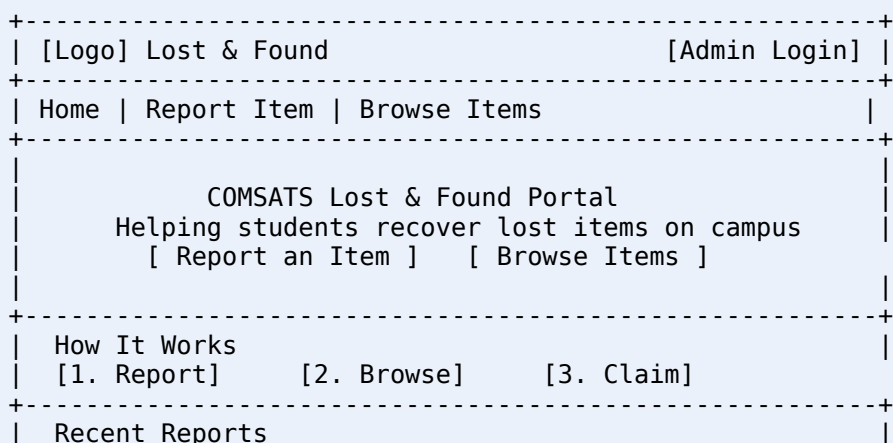
2. Sitemap

The sitemap below shows how the portal's pages connect. Every page shares the same top navigation bar, and the Admin Dashboard is reached only after a successful login through the Admin Login modal.



3. Home Page Wireframe

A rough layout plan for the Home page was sketched before coding, identifying the position of the header, navigation, hero section, informational cards, and the recent-reports feed:



```

| [Item card] [Item card] [Item card] |
+-----+
| (c) 2026 COMSATS Lost & Found Portal |
+-----+

```

4. Development Environment

The following tools and technologies made up the development environment used for this project:

Tool	Purpose
Code Editor	Visual Studio Code — used to write the HTML, CSS, and JavaScript
Browser	Google Chrome — used to run and test the site, including DevTools for inspecting elements and checking the console
Local Server	XAMPP — used to serve the project locally over HTTP instead of opening files directly from disk
Operating System	Windows — the lab operating environment
Deployment Platform	Vercel — used to host the finished portal for live, public access

5. Tiered Architecture of the Portal

The same Lost & Found project was examined under three architecture models, as required by the lab manual.

5.1 One-Tier Architecture

In the one-tier setup, the browser alone handles everything: rendering the HTML, applying the CSS, and running the JavaScript directly from local files (file://), with no server involved and no data able to persist between visits.

```

[ Browser ]
  |
HTML + CSS + JavaScript
(load directly from disk, no server, no persistence)

```

5.2 Two-Tier Architecture

In the two-tier setup, the browser (client) sends an HTTP request to a web server, which responds with the requested page and data. This is the model demonstrated when the project was served locally through XAMPP at `http://localhost/`, and again in production where the browser talks directly to the Vercel-hosted server.

```

[ Browser (Client) ] --HTTP Request--> [ Web Server ]
[ Browser (Client) ] <--HTTP Response--- [ Web Server ]

```

5.3 N-Tier Architecture

For an N-tier version of the portal, responsibilities would be split into separate logical layers instead of one server handling everything:

```

[Browser] -> [Presentation] -> [Business Logic] -> [Data Access] -> [Database]

```

```

Presentation    : renders Home / Report Item / Browse Items / Admin pages
Business Logic  : validates report form fields, checks admin credentials,
                  applies claim approval / resolve rules
Data Access     : reads/writes item, claim, and admin-account records
Database        : stores reported items, their status, and claim records

```

A tier is a logical separation of responsibility, not a fixed number of layers — N-tier simply means the system is divided into more than two cooperating layers.

6. Local Server Testing with XAMPP

To fulfil Task 4 of the lab manual, the project was placed in XAMPP's local web-server environment and the Apache service was started. The site was then accessed through `http://localhost/` instead of opening the files directly, confirming a genuine HTTP client-server exchange between the browser and the local web server rather than a simple local file load.

All pages — Home, Report Item, Browse Items, and the Admin Login / Admin Dashboard flow — were verified to work correctly when served this way, including form submission, search and filtering, and admin authentication.

7. Page Screenshots

The screenshots below were captured from the live, deployed version of the portal (see Section 8), and show the same pages and flows that were verified locally under XAMPP.

7.1 Home Page

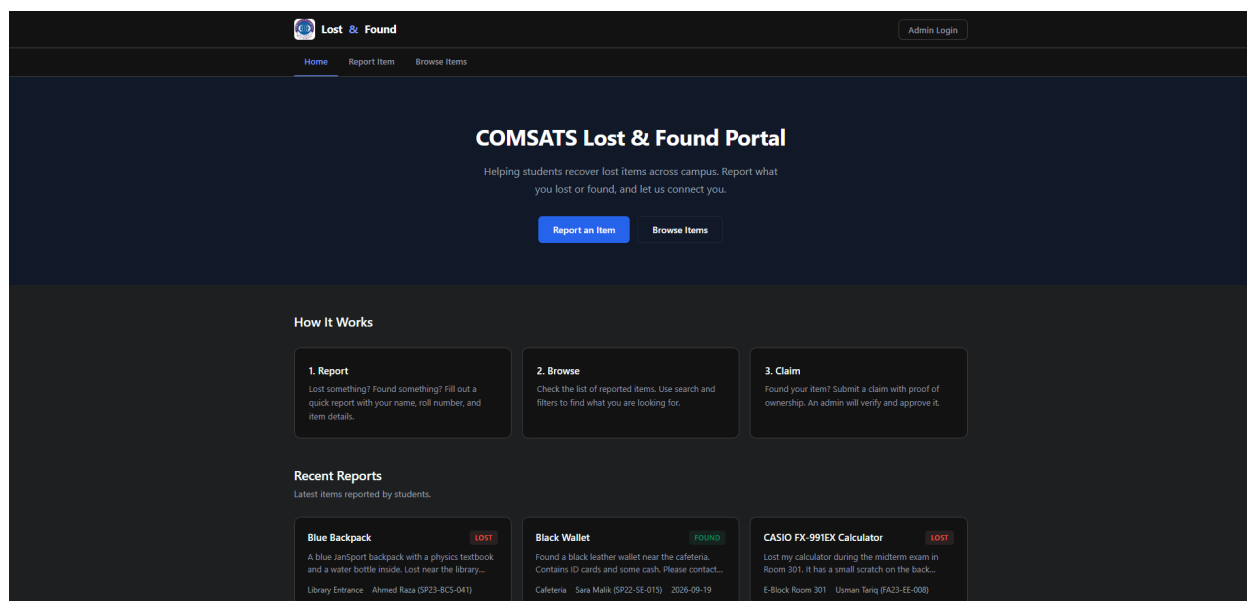


Figure 1: Home page — hero section, “How It Works” steps, and Recent Reports feed.

7.2 Admin Login

The screenshot shows the 'Admin Login' modal on the COMSATS Lost & Found Portal. The modal is titled 'Admin Login' and contains the following fields and buttons:

- Username:** A text input field with the placeholder text 'Enter admin username'.
- Password:** A text input field with the placeholder text 'Enter admin password'.
- Login:** A blue button to submit the login credentials.

The background of the portal shows the 'COMSATS Lost & Found Portal' header, a navigation bar with 'Home', 'Report Item', and 'Browse Items', and a 'How It Works' section with '1. Report' and '2. Claim' steps. Below this is a 'Recent Reports' section listing items like 'Blue Backpack', 'Black Wallet', and 'CASIO FX-991EX Calculator'.

Figure 2: Admin Login modal, used to authenticate before reaching the Admin Dashboard.

7.3 Report an Item

The screenshot shows the 'Report an Item' form on the COMSATS Lost & Found Portal. The form is titled 'Report an Item' and contains the following fields and buttons:

- Your Name *:** A text input field with the placeholder text 'e.g., Ahmed Raza'.
- Roll Number *:** A text input field with the placeholder text 'e.g., SP23-BCS-041'.
- Item Type *:** A dropdown menu with the placeholder text 'Select type'.
- Location *:** A text input field with the placeholder text 'e.g., Library, C-Block'.
- Item Title *:** A text input field with the placeholder text 'e.g., Blue Backpack, Student ID Card'.
- Description *:** A text area with the placeholder text 'Describe the item: color, brand, distinguishing features, when/where it was lost or found...'.
- Contact Email *:** A text input field with the placeholder text 'your@comsats.edu.pk'.
- Submit Report:** A blue button to submit the report.

The background of the portal shows the 'COMSATS Lost & Found Portal' header, a navigation bar with 'Home', 'Report Item', and 'Browse Items', and a 'How It Works' section with '1. Report' and '2. Claim' steps. Below this is a 'Recent Reports' section listing items like 'Blue Backpack', 'Black Wallet', and 'CASIO FX-991EX Calculator'.

Figure 3: Report Item form, used by students to submit a lost or found item.

7.4 Browse Items

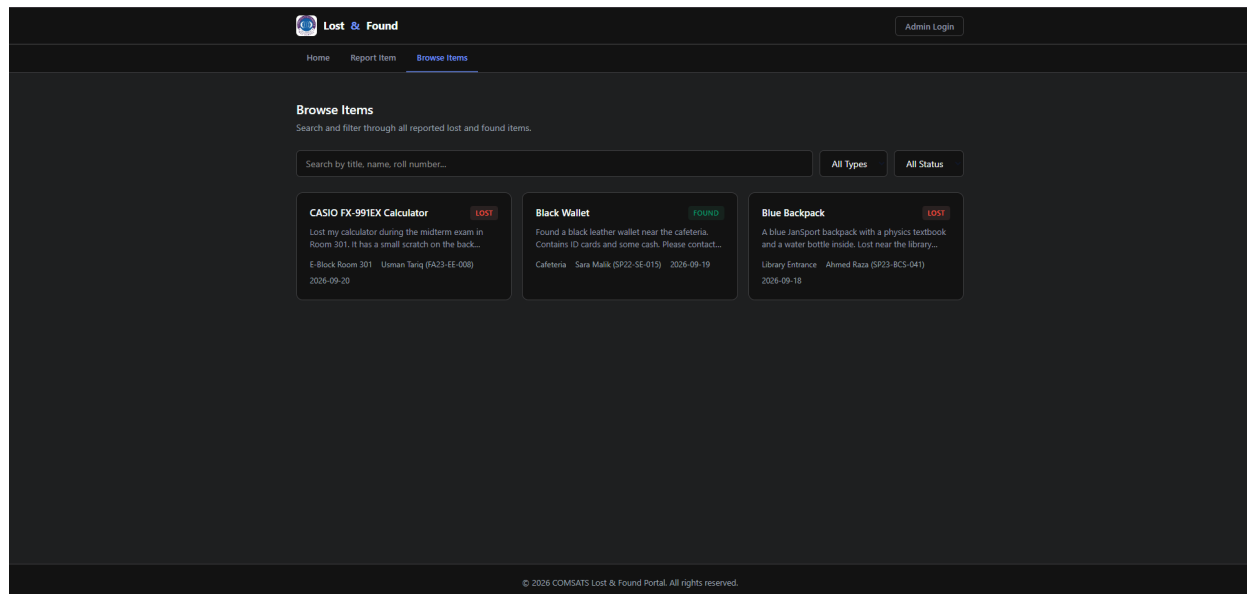


Figure 4: Browse Items page, with search and Type / Status filters over reported items.

7.5 Admin Dashboard

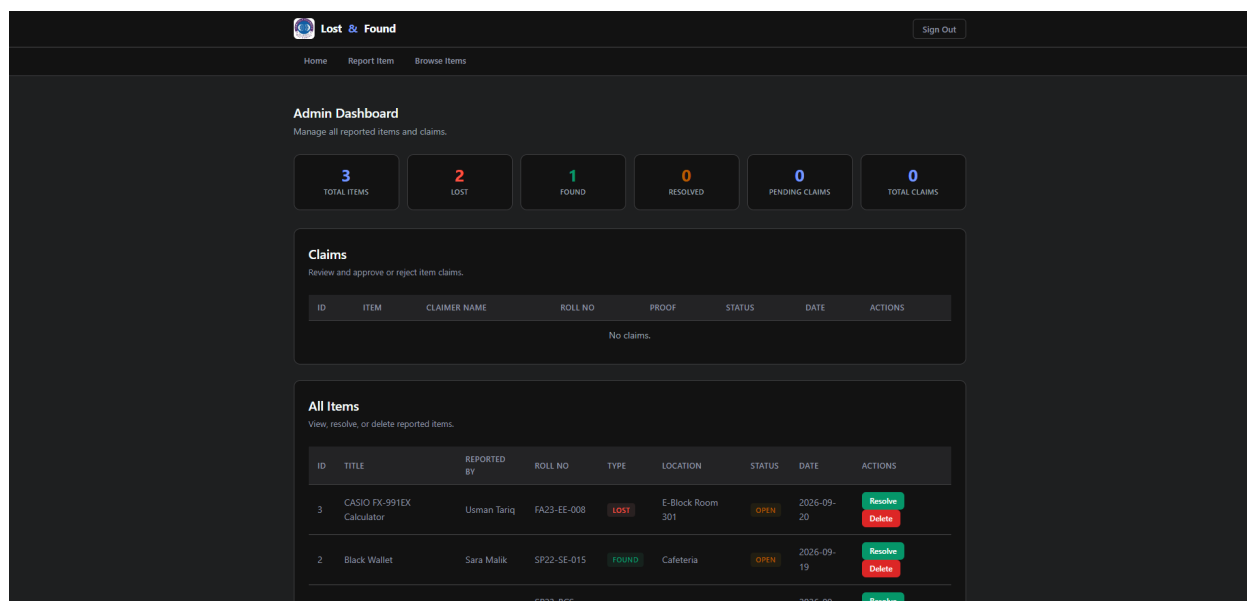


Figure 5: Admin Dashboard — statistics, claims review, and item management (resolve / delete).

8. Live Website Link

The portal was deployed using Vercel and is publicly available at:

<https://webtech-lab1-eosin.vercel.app>